

K.A.L.A KCSE 2025 PREDICTION

(KENYA HIGH-ALLIANCE -LENANA-ALLIANCE GIRLS)

121/2

MATHEMATICS

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES:

- Write your **name**, **admission number** and write **date** of examination in the spaces provided
- The paper contains **two** sections. Section I and Section II.
- Answer **ALL** the questions in section I and any **five** questions in section II.
- Answers and working **must** be written on the question paper in the spaces provided below each question.
- Show all steps in your calculations below each question.
- Marks may be given for correct working even if the answer is wrong.
- Non programmable silent electronic calculators and KNEC mathematical table may be used, except where stated otherwise.

FOR EXAMINERS USE ONLY

SECTION I

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL
Marks																	

SECTION II

Question	17	18	19	20	21	22	23	24	TOTAL
Marks									

GRAND

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SECTION I: 50 MARKS

ANSWER ALL QUESTIONS IN THIS SECTION

1. Solve for x in the equation

$$2\sin^2 x - 1 = \cos^2 x + \sin x \text{ for } 0 \leq x \leq 360 \quad (3 \text{ Marks})$$

2. (a) Expand $\left(1 + \frac{3}{x}\right)^5$ upto the fifth term (2 Marks)

(b) Hence use your expansion to evaluate the value of $(2.5)^5$ to 3 d.p. (2 Marks)

3. Make p the subject of the formula

(3 Marks)

$$E + m = m + \sqrt{\frac{p - 3u}{y - 3mp}}$$

4. A object A of area 10cm^2 is mapped onto its image B of area 60cm^2 by a transformation whose matrix is given by

$P = \begin{pmatrix} x & 4 \\ 3 & x + 3 \end{pmatrix}$. Find the possible value of

(3 Marks)

5. A variable Z varies directly as the square of X and inversely as the square root of Y. Find the percentage change in Z if X increased by 20% and Y decreased by 19% **(3 Marks)**

6. A circle whose equation is $(x - 1)^2 + (y - k)^2 = 10$ passes through point (2,5). Find the coordinates of the two possible centres of the circle. **(3marks)**

7. Juma deposited Sh. 45 000 in a bank which paid interest at 12% p.a compounded monthly. Calculate the amount of interest after 3 years **(3 Marks)**

8. (a) Find the inverse of the matrix $\begin{pmatrix} 4 & 3 \\ 3 & 5 \end{pmatrix}$ **(1 Mark)**

- (b) Hence solve the simultaneous equation below using matrix method **(3 Marks)**

$$4x + 3y = 6$$

$$5y + 3x - 5 = 0$$

9.

10. Given that $P=4+\sqrt{2}$ and $Q=2+\sqrt{2}$ and that $\frac{P}{Q}=a+b\sqrt{c}$, where a, b and c are constants, find the values of a, b and c. **(3 marks)**

11. Find the value of x that satisfies the equation:

$$\log_3(x + 24) - 2 = \log_3(9 - 2x) \quad \textbf{(3 marks)}$$

12. Under a shear with x-axis invariant the point (3,2) is mapped onto (-2,2). Find the image of point (4,4) under the same transformation. **(3marks)**

13. The data below shows the marks obtained by ten students in a test.

71, 55, 69, 45, 65, 57, 71, 82, 55, 50

Find the quartile deviations.

(3 marks)

14. A bag contains 4 green, 9 purple and 18 blue balls. The balls are identical except for the colour.

Two balls are picked at random, one at a time without replacement, find the probability that, a blue ball and a green ball are picked.

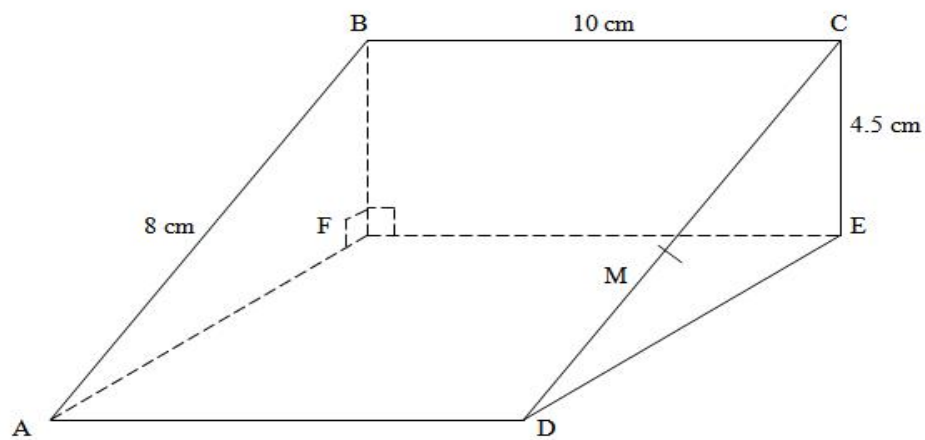
(3 marks)

15. Find the equation of the tangent to the curve $y = x^2 + 2x + 3$ at the point where the gradient is equal to 4.

(3 marks)

(3 marks)

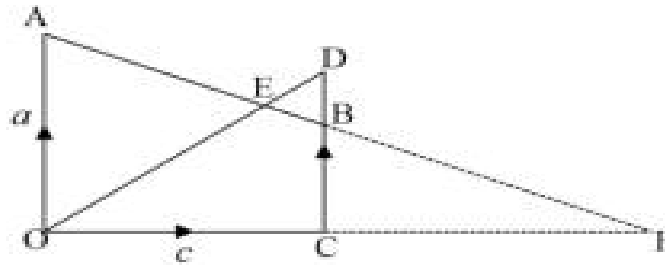
(3marks)



SECTION II: 50 MARKS

ANSWER ANY 5 QUESTIONS IN THIS SECTION

18. In the figure below, $\mathbf{OA} = \tilde{a}$ and $\mathbf{OC} = \tilde{c}$. $\mathbf{CB} = \frac{2}{3}\mathbf{OA}$ and B divides CD in the ratio 3:1



(a) Express the following vectors in terms of **a** and **c** only:

(i) **AB** (1 mark)

(ii) **OD** (2 marks)

(b) Given that $\mathbf{OE} = h\mathbf{OD}$ and $\mathbf{AE} = k\mathbf{AB}$ where h and k are scalars express $\mathbf{OE}$ in two different ways hence find the scalars h and k . (5 marks)

(c) If $\mathbf{OC}$ produced meets $\mathbf{AB}$ produced at $\mathbf{F}$, find $\mathbf{OF}$. (3 marks)

19. (a) The 5th term of an AP is 82 and the 12th term is 103. find:

(i) the first term and common difference. **(3marks)**

(ii) the sum of the 21 terms. **(2 marks)**

(b) A stair case was built such that each subsequent stair has a uniform difference in height. The height of the 6th stair from the horizontal floor was 85 cm and the height of the 10th stair was 145. Calculate the height of the 1st stair and the uniform difference in height of the stairs.

(3 marks)

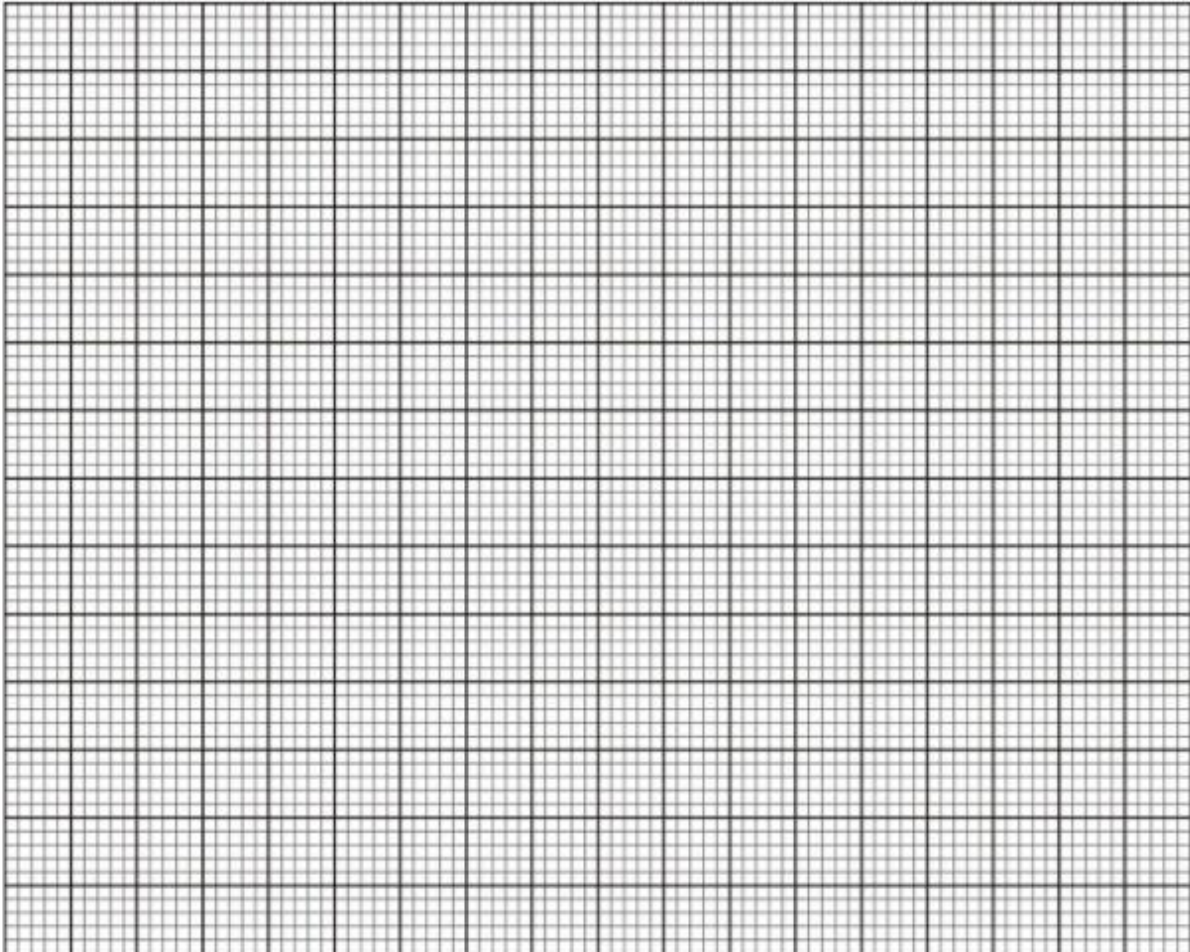
(c) During the construction of the staircase, each step was supported by a vertical piece of timber. If the staircase has 11 stairs, calculate the total length of timber used. **(2 marks)**

19.(a) Complete the table below given that $y = -x^2 + x + 12$ for $-4 \leq x \leq 5$ **(2 marks)**

X	-4	-3	-2	-1	0	1	2	3	4	5
Y	8				12	12				

(b) On the grid provided below, draw the graph of $y = -x^2 + x + 12$ for $-4 \leq x \leq 5$

Scale: 1 cm represents 1 unit on the x – axis and 1 cm represents 2 units on the y – axis. **(4 marks)**



(c) Using your graph, to Solve the equation $x^2 - 3x - 10 = 0$ **(2 marks)**

ii. State the range of values of x for which $-x^2 + x + 12 \geq 0$ **(1 mark)**

(d) Estimate the coordinates of the turning point of the curve $y = -x^2 + x + 12$ **(1 mark)**

20. An aircraft leaves town $P(30^{\circ}S, 17^{\circ}E)$ and moves directly to town $Q(60^{\circ}N, 17^{\circ}E)$. It then moved at an average speed of 300 knots for 8 hours westwards to town R. Determine;

(a) The distance PQ in nautical miles. **(3 marks)**

(b) The position of town R. **(3 marks)**

(c) The local time at R if the local time at Q is 3:15 p.m **(2 marks)**

(d) The total distance moved from P to R in km. Take $1\text{nm}=1.853\text{ km}$ **(2 marks)**

21. Using a ruler and compasses only.

(i) Construct a parallelogram ABCD such that $AB = 10\text{cm}$, $BC = 7\text{cm}$ and $\angle ABC = 105^\circ$.

(3 marks)

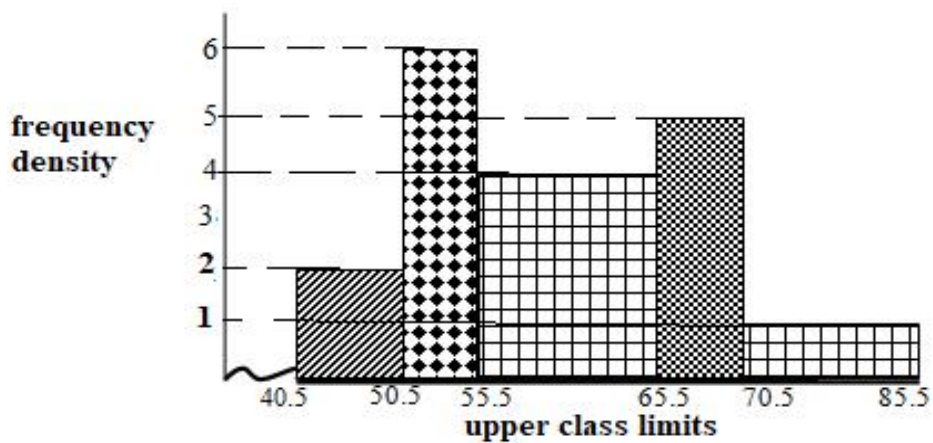
(ii) Construct the loci of P and Q within the parallelogram such that $AP \leq 4\text{cm}$ and $BQ \leq 6\text{cm}$

(3 marks)

(iii) Calculate the area within the parallelogram but outside regions bounded by the loci of P and Q.

(4 marks)

22. Below is the histogram representing marks obtained in Mathematics test



(a) Develop a frequency distribution table for the data (3 Marks)

(b) Using an assumed mean of 60.5 find the mean. (3 marks)

(c) Calculate Standard deviation. (4marks)

23. (a) Use the mid ordinate rule with 5 strips to estimate the area bounded by the curve $y = x^2 - 3x - 4$, $x = -2$, $x = 3$ and x - axis **(3 marks)**

(b) Calculate the exact area above **(5 marks)**

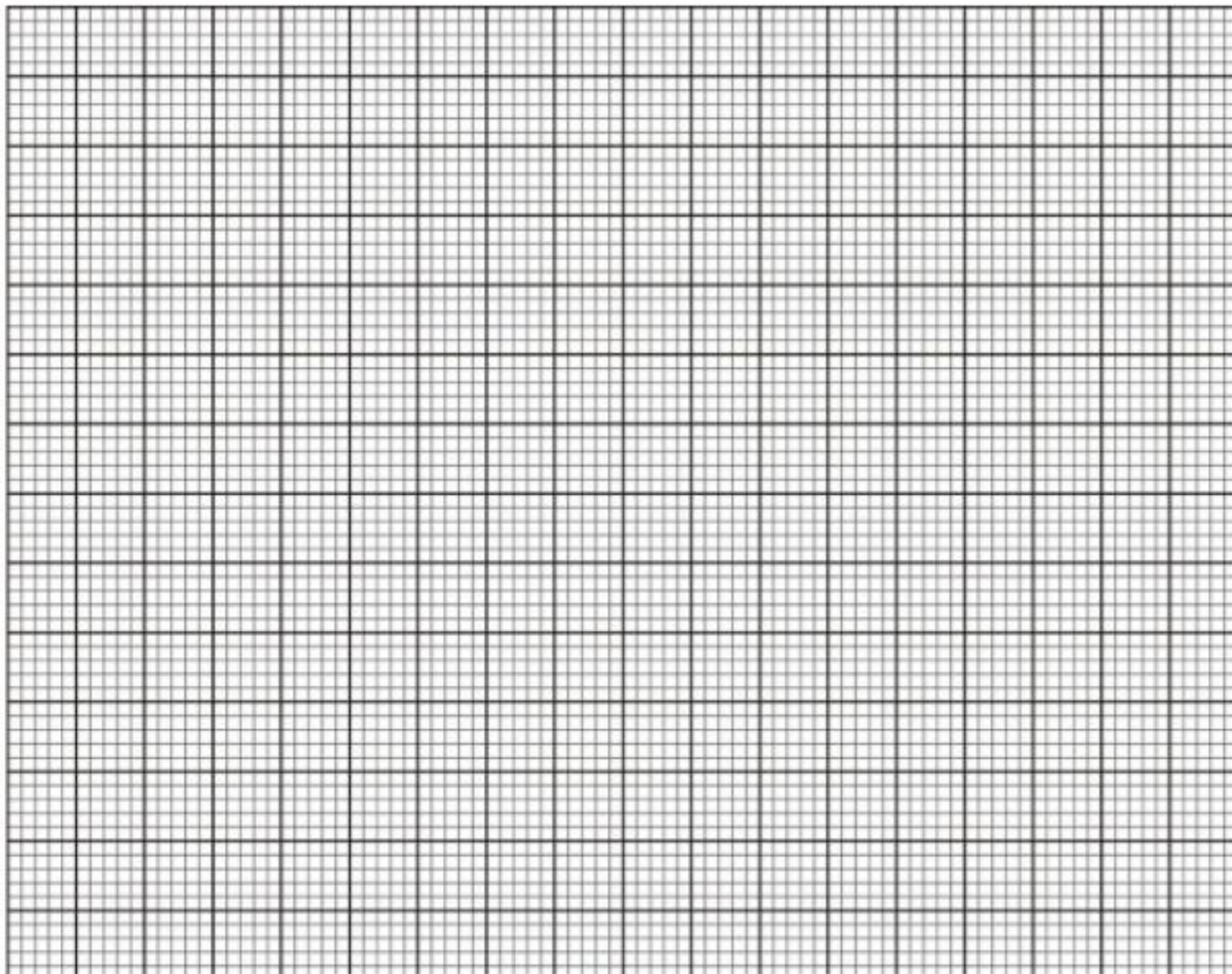
(c) Find the percentage error involve in using the mid-ordinate role. **(2 marks)**

24.

25. Eldoret Airport is planning to build a fire fighting plant on a space of 250m^2 . Two types of machines are to be installed, machine x which occupies a space of 5m^2 and machine Y which occupies 10m^2 . The airport can have a maximum of 40 machines at a time. At most 15 machines of type Y are used at any given time.

a) Write down three inequalities other than $x > 0$, and $y > 0$. **(3marks)**

b) On the grid below, show the region satisfying the given conditions. **(4marks)**



- c) The profit from a type x machine is Ksh 1000 and that of type y is Ksh 4000. Use the graph to obtain the number of machines of each type that should be installed to obtain maximum profit. Calculate the maximum profit.
(3marks)